Statistical Analysis of Exoplanet Discovery Data:

Comprehensive Review of Kepler and TESS Datasets and Habitable Zone Probabilities

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Abstract

The discovery and characterization of exoplanets has revolutionized our understanding of planetary systems. Large-scale surveys conducted by NASA's Kepler and TESS missions have generated vast datasets containing thousands of confirmed and candidate exoplanets, enabling detailed statistical studies of planetary populations. This paper presents a literature-based and conceptual review of statistical methods applied to exoplanet discovery data, focusing on identifying occurrence rates, population patterns, and probabilities of planets residing in habitable zones. Emphasis is placed on methodologies such as transit probability calculations, occurrence rate estimation, and statistical modeling of planetary radii and orbital periods. By synthesizing insights from the Kepler and TESS datasets, this review aims to provide a comprehensive understanding of trends in exoplanet demographics and implications for the search for potentially habitable worlds.

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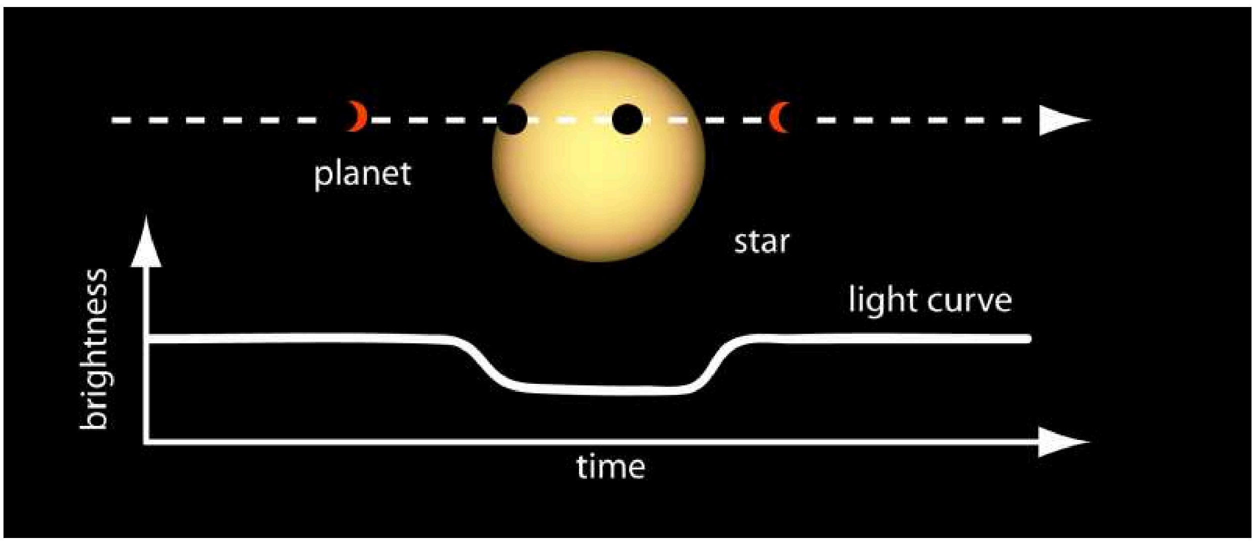
1. Introduction

The study of exoplanets has progressed rapidly since the first confirmed detections in the 1990s. With the launch of NASA's Kepler mission in 2009 and the TESS mission in 2018, astronomers now have access to high-precision photometric datasets that allow the detection of planetary transits across host stars (Borucki et al., 2010; Ricker et al., 2015). These datasets provide an unprecedented opportunity to conduct statistical analyses, identifying patterns in planetary occurrence, distributions of sizes and orbital periods, and the likelihood of planets existing in habitable zones—regions around stars where liquid water may exist.

Statistical analysis of exoplanet data is crucial for several reasons. First, it allows researchers to correct for observational biases, such as geometric transit probabilities and detection efficiency. Second, it enables the derivation of planetary occurrence rates, which are essential for understanding planet formation and evolution. Third, it supports the estimation of the frequency of potentially habitable planets, which has profound implications for astrobiology and the search for extraterrestrial life (Petigura et al., 2013).

This paper presents a comprehensive review of statistical methodologies applied to Kepler and TESS datasets, emphasizing techniques for analyzing large-scale exoplanet populations, estimating occurrence rates, and assessing habitable zone probabilities. The review integrates findings from both missions and discusses limitations, uncertainties, and avenues for future research.



2. Foundations of Exoplanet Detection and Statistical Methods

2.1 Transit Method of Exoplanet Detection

The primary method used by Kepler and TESS is the transit technique, which detects periodic dips in stellar brightness caused by planets crossing in front of their host stars. The probability that a planet transits depends on its orbital radius and the radius of the host star (Seager & Mallén-Ornelas, 2003). Statistical corrections for this geometric bias are essential to infer the true population of planets.

2.2 Statistical Frameworks in Exoplanet Analysis

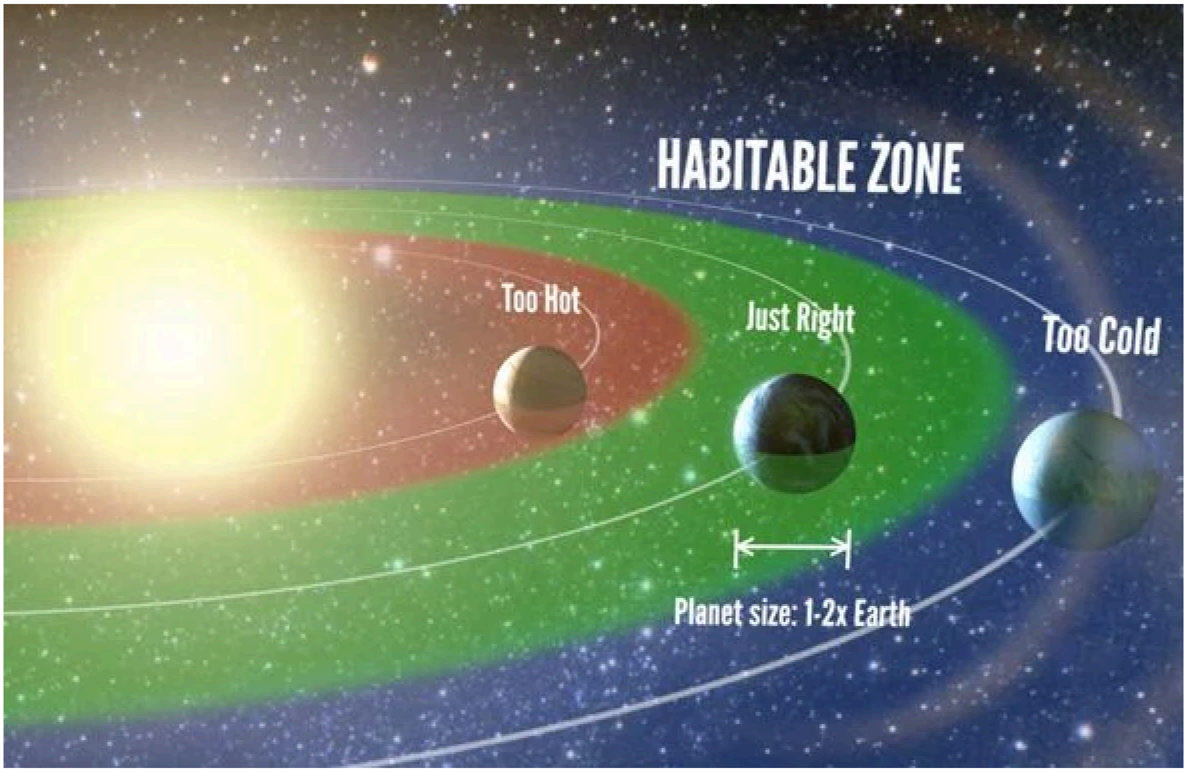
Statistical analysis of exoplanet populations involves several key components:

1. **Occurrence Rate Estimation**: The number of planets per star within a specific parameter range (e.g., radius, orbital period) is estimated using techniques such as inverse detection efficiency weighting (Howard et al., 2012).

2. **Bias Correction**: Detection efficiency, signal-to-noise thresholds, and survey completeness are incorporated into models to correct raw counts (Christiansen et al., 2015).

3. **Probabilistic Modeling**: Bayesian frameworks are commonly used to account for uncertainties in planet parameters and to derive posterior distributions of planetary occurrence (Foreman-Mackey et al., 2014).

4. **Habitable Zone Calculations**: Habitable zone (HZ) boundaries are defined based on stellar luminosity and effective temperature. Planets are classified probabilistically based on their likelihood of residing within these limits (Kopparapu et al., 2013).



2.3 Importance of Statistical Methods

Without statistical modeling, analyses of exoplanet populations would be biased toward planets with short orbital periods, large radii, and bright host stars. Statistical methods allow astronomers to make inferences about the true underlying population, including Earth-sized planets in the habitable zones of Sun-like stars.

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3. Kepler and TESS Mission Datasets

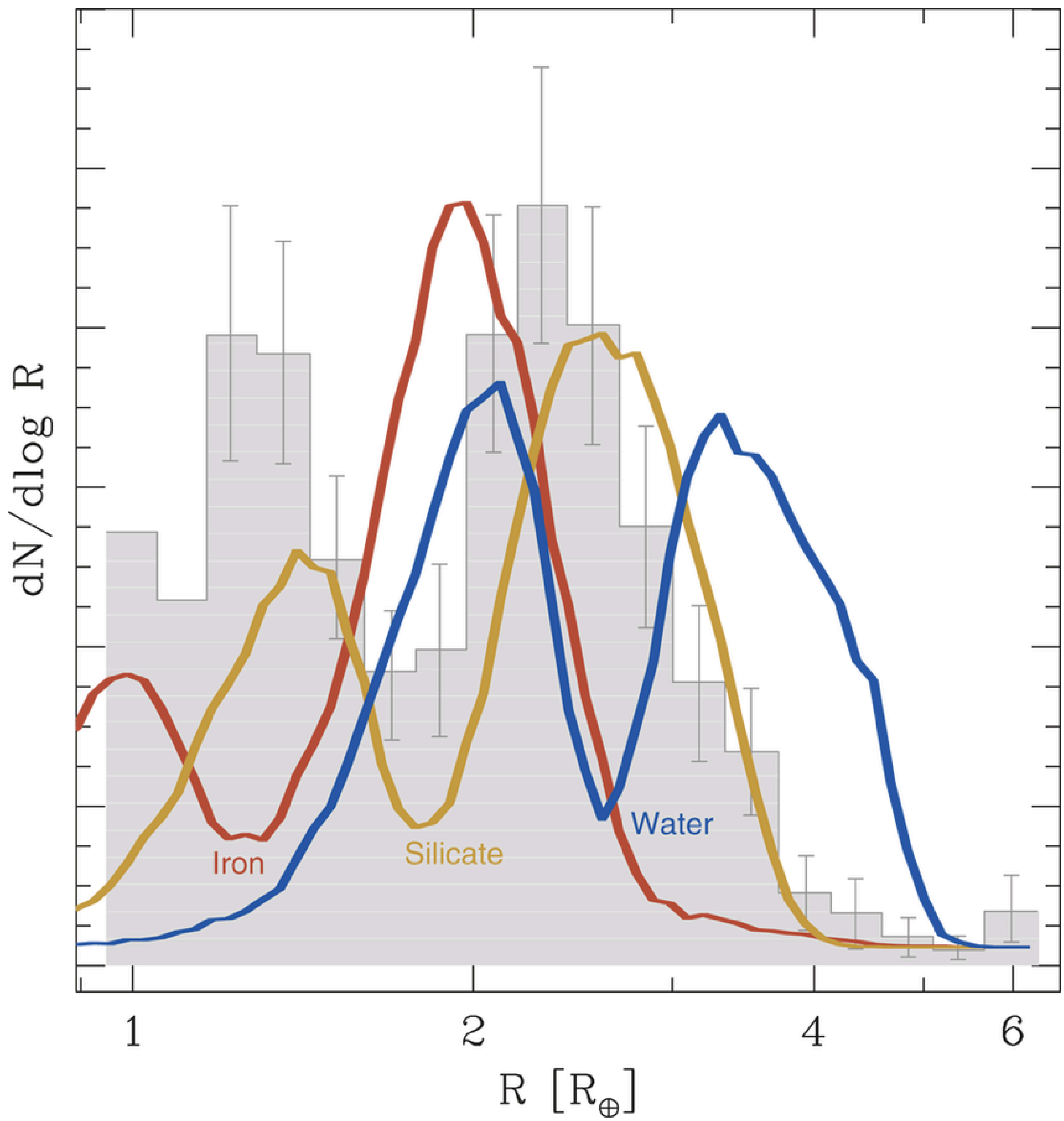
3.1 Kepler Mission

The Kepler space telescope monitored over 150,000 stars in a fixed field for approximately four years, detecting thousands of exoplanet candidates via the transit method (Borucki et al., 2010). Key datasets include:

- **Kepler Object of Interest (KOI) Catalog**: Contains candidate planets with measured radii, orbital periods, and transit depths.

- **Confirmed Planets Catalog**: Includes planets verified via follow-up observations or statistical validation.

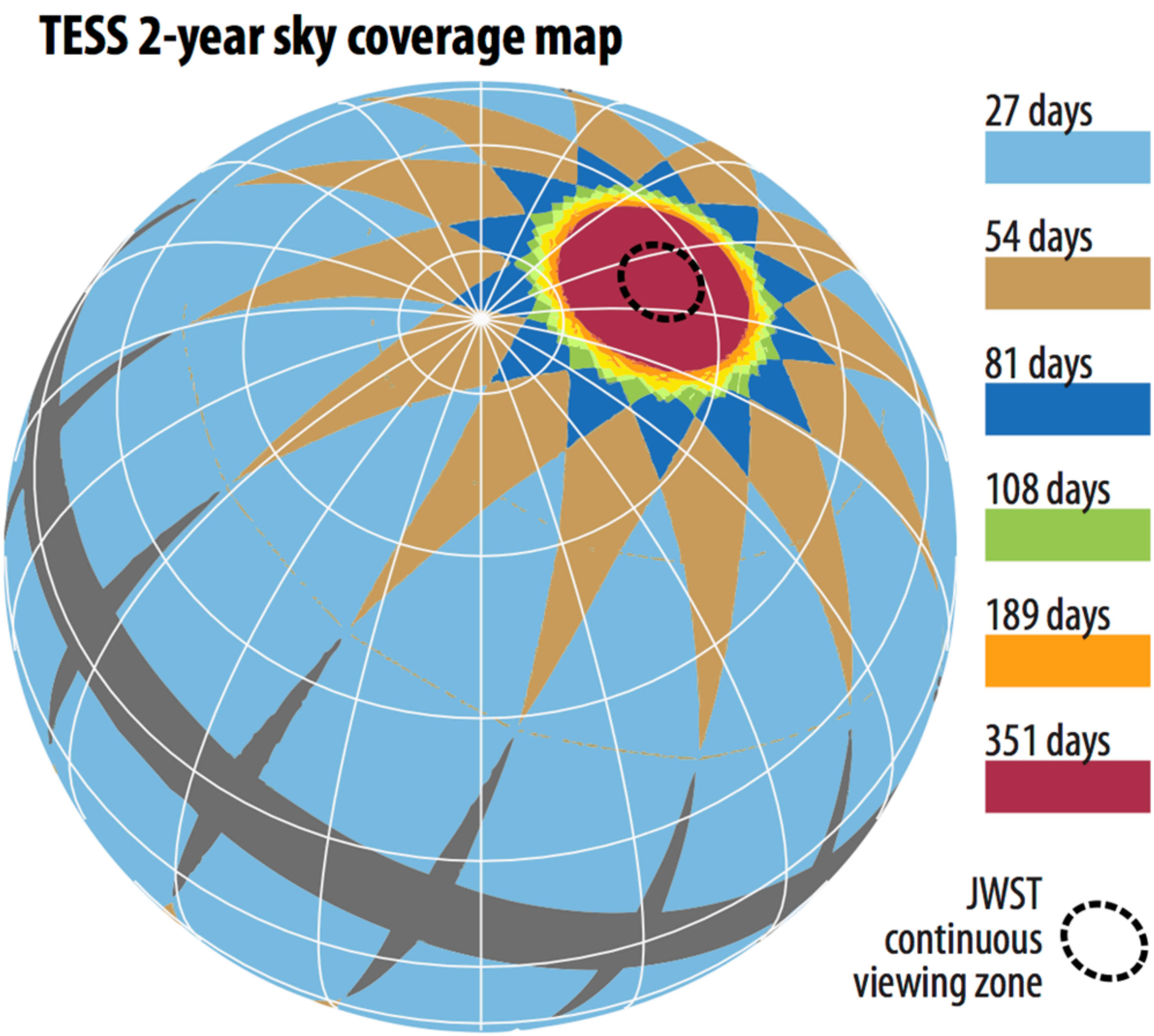
Kepler's high photometric precision enables the detection of small, Earth-sized planets and provides the foundation for statistical studies of planet occurrence rates.



Distribution of Kepler planet radii vs orbital periods

3.2 TESS Mission

TESS conducts an all-sky survey, focusing on bright, nearby stars to facilitate follow-up observations. Unlike Kepler, TESS observes each sector for ~27 days, detecting shorter-period planets with high signal-to-noise ratios (Ricker et al., 2015). TESS provides complementary datasets, expanding the range of host star types and planetary systems available for analysis.



3.3 Data Preprocessing and Quality Control

Before statistical analysis, datasets undergo rigorous preprocessing, including:

- Removal of false positives (stellar variability, eclipsing binaries)
- Detrending of light curves to remove instrumental and astrophysical noise
- Verification of planetary parameters (radius, orbital period, semi-major axis)

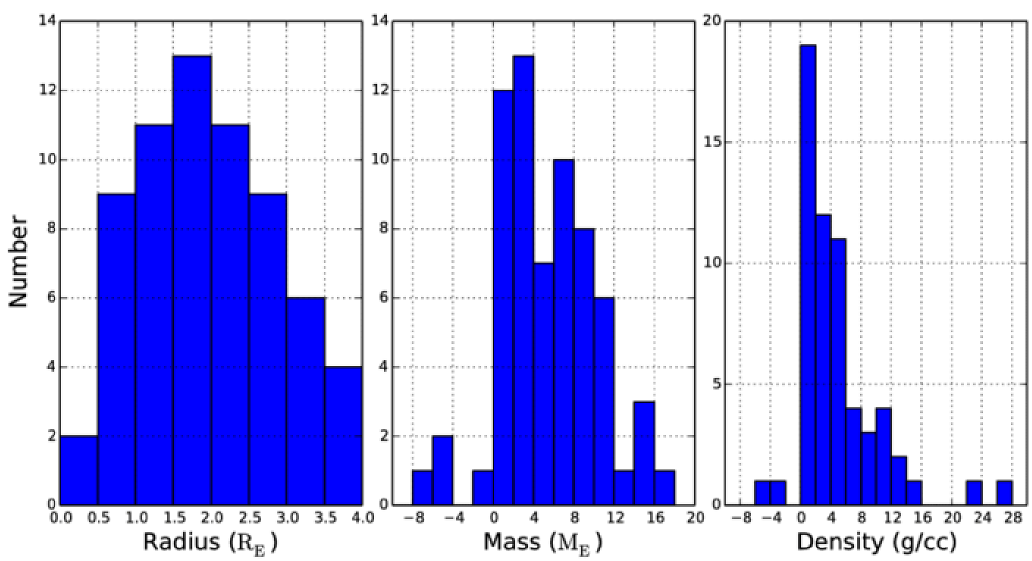
Such preprocessing is critical to ensure reliable statistical inferences.

4. Statistical Patterns in Planetary Populations

4.1 Planetary Radius Distribution

Analysis of Kepler and TESS datasets reveals distinct patterns in planetary radii. Kepler data shows a bimodal distribution, with peaks at ~1.3 Earth radii (super-Earths) and ~2.4 Earth radii (sub-Neptunes), often referred to as the "radius gap" (Fulton et al., 2017). This gap suggests that atmospheric loss processes, such as photoevaporation, play a role in shaping the final sizes of planets.

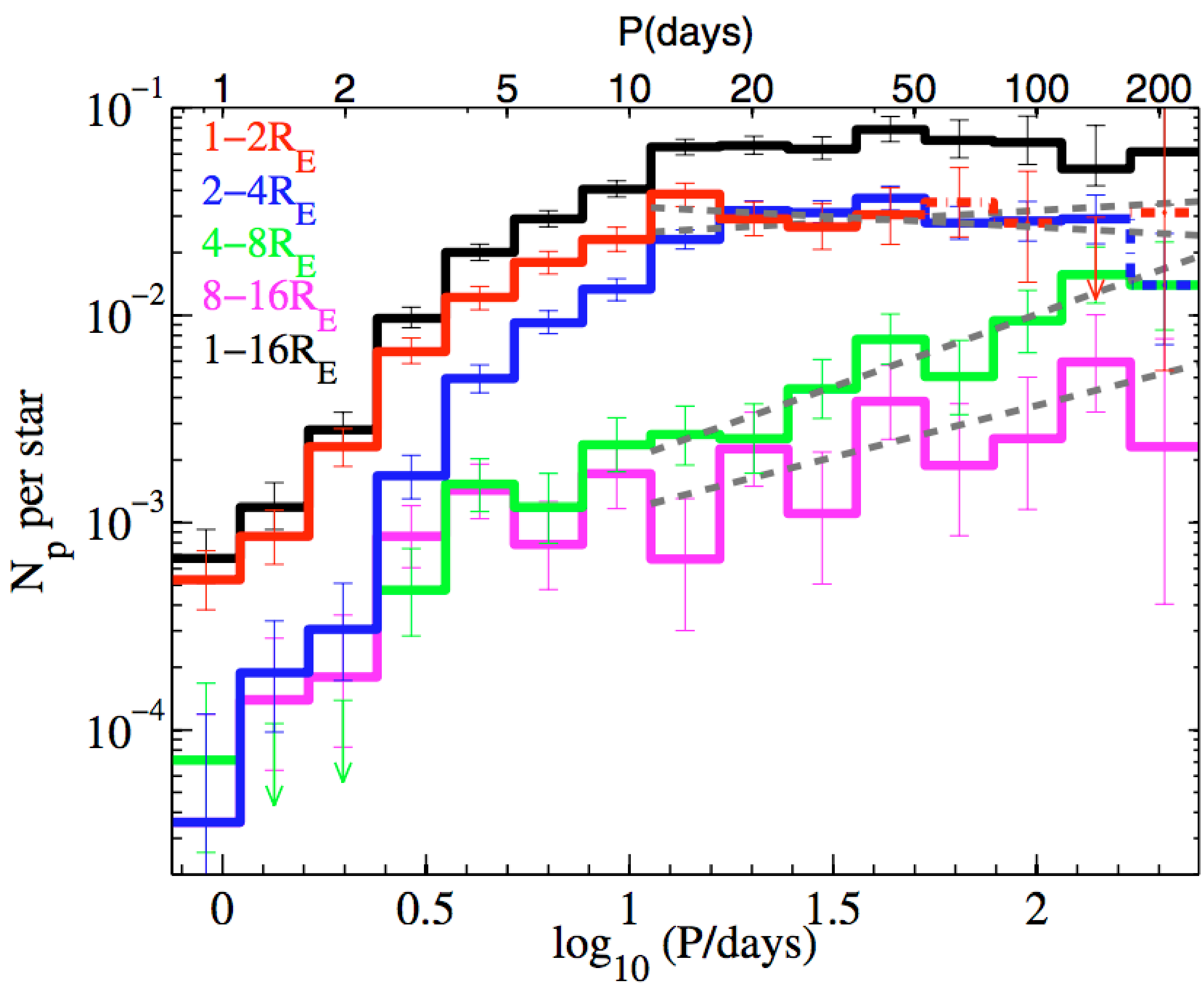
TESS, while observing shorter-period planets, confirms the general trend but is more sensitive to larger planets due to its shorter observation duration and smaller photometric depth for small planets. Statistical modeling of these distributions involves fitting probability density functions and correcting for detection biases.



4.2 Orbital Period Distribution

Exoplanet orbital periods also exhibit characteristic patterns. Both Kepler and TESS datasets show that small planets are more common at short orbital periods (<50 days), with a steep decline in frequency for longer periods due to observational incompleteness (Howard et al., 2012). Hot Jupiters are rare but more easily detected due to their large transit signals.

Statistical analyses often use kernel density estimation or Bayesian hierarchical modeling to derive occurrence rates as functions of orbital period and radius simultaneously. These models allow for extrapolation to planets not yet detected, improving estimates of the underlying population.



Orbital period vs planet occurrence rate from Kepler data

4.3 Multiplicity and System Architecture

Kepler revealed that multi-planet systems are common, particularly among smaller planets. Statistical studies indicate that planets in multi-planet systems tend to have low mutual inclinations and relatively uniform spacing, implying dynamically stable configurations (Lissauer et al., 2011). TESS, though observing fewer long-period planets, is beginning to detect similar trends among nearby bright systems.

Understanding system architecture requires probabilistic modeling of transit probability, accounting for observational biases, and often involves Monte Carlo simulations to infer the frequency of undetected planets.

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5. Habitable Zone Probabilities and Candidate Planets

5.1 Definition of the Habitable Zone

The habitable zone (HZ) is the circumstellar region where conditions allow for liquid water on a planet's surface. Calculating HZ boundaries depends on the host star's luminosity, temperature, and spectral type (Kopparapu et al., 2013). The inner edge is limited by runaway greenhouse effects, while the outer edge is constrained by maximum greenhouse warming.

5.2 Statistical Estimation of Habitable Planets

Determining the probability that a planet resides in the HZ involves comparing its orbital parameters to the computed HZ boundaries. Bayesian inference and Monte Carlo sampling are commonly applied to incorporate uncertainties in stellar and planetary parameters (Foreman-Mackey et al., 2014).

Current statistical analyses suggest that the occurrence rate of Earth-sized planets in the HZ around Sun-like stars, denoted as $\eta_{\oplus}$\eta_\oplus$\eta_{\oplus}$, is roughly 0.1–0.2 based on Kepler data (Petigura et al., 2013). TESS is expected to refine these estimates by targeting bright nearby stars that facilitate follow-up observations.

5.3 Candidate Selection and Validation

Candidate planets in the HZ are initially identified through transit signals and orbital modeling. False positives, such as eclipsing binaries or stellar variability, are statistically removed using vetting pipelines and validation algorithms. Confirmed HZ planets represent prime targets for future atmospheric characterization with telescopes such as the James Webb Space Telescope (JWST) and ground-based observatories.

6. Comparative Analysis of Kepler vs TESS Data

6.1 Mission Characteristics

Kepler and TESS differ in mission design:

- **Kepler**: Focused on a fixed field of ~150,000 stars for ~4 years, sensitive to small, long-period planets.
- **TESS**: Conducts an all-sky survey, observing each sector for ~27 days, optimized for short-period planets around bright stars.

These differences lead to complementary datasets that, when combined, provide a broader statistical view of exoplanet populations.

6.2 Differences in Detection Biases

Kepler's long-duration observations allow the detection of multiple transits for planets with orbital periods up to several hundred days, whereas TESS preferentially detects shorter-period planets. Statistical models must account for these biases to compare occurrence rates accurately.

6.3 Integrated Statistical Insights

By integrating Kepler and TESS data, researchers can derive more robust occurrence rates across a wide range of stellar types and orbital periods. For instance:

- The fraction of stars hosting small, short-period planets is well-constrained.
- The frequency of potentially habitable Earth-sized planets around Sun-like stars is statistically refined.
- Trends in planetary multiplicity, system architecture, and radius distribution are corroborated across both datasets.

7. Limitations and Observational Biases

7.1 Detection Sensitivity and Completeness

One of the main limitations in exoplanet statistical studies is **detection sensitivity**. The ability of Kepler and TESS to detect planets depends on the planet's size, orbital period, and host star brightness. Small planets or planets around faint stars are often undetectable, leading to incompleteness in the observed sample (Christiansen et al., 2015). Statistical models attempt to correct for this by weighting planets according to their detection probability, but uncertainties remain, especially for longer-period planets that produce fewer transits.

7.2 False Positives and Vetting Limitations

Transit surveys are prone to false positives caused by eclipsing binaries, stellar variability, or instrumental noise. While automated vetting pipelines reduce contamination, some false positives may remain, affecting statistical inferences. Confirmation through radial velocity measurements or high-resolution imaging is often limited by observational resources, particularly for faint or distant stars (Coughlin et al., 2016).

7.3 Stellar Parameter Uncertainties

Accurate determination of planetary parameters relies on precise knowledge of host star properties, such as radius, mass, and luminosity. Uncertainties in stellar characterization propagate into estimates of planet radius, orbital semi-major axis, and habitable zone placement. While missions like Gaia have improved stellar parameter accuracy, residual errors still affect occurrence rate calculations and habitable planet probabilities (Berger et al., 2020).

7.4 Survey Duration and Orbital Bias

Kepler's fixed field allowed the detection of long-period planets, whereas TESS's shorter observation duration biases detections toward short-period planets. Statistical models must account for this orbital bias, but extrapolation to long-period planets remains uncertain, affecting estimates of habitable zone planet frequencies.

8. Future Directions and Surveys

8.1 Upcoming Space Missions

Future missions promise to expand statistical analyses of exoplanet populations:

- **PLATO (ESA)**: Will focus on bright stars, long-duration monitoring, and small planets in habitable zones.
- **JWST**: Will enable atmospheric characterization of transiting exoplanets, particularly HZ candidates, refining habitability assessments.
- **Roman Space Telescope**: Will provide microlensing surveys to probe distant planets, complementing transit statistics.

These missions will increase the sample size, improve stellar characterization, and allow direct tests of occurrence rate predictions derived from Kepler and TESS data.

8.2 Improved Statistical Methods

Advancements in statistical methodology are essential to extract maximal information from exoplanet surveys:

- **Hierarchical Bayesian Models**: Enable simultaneous inference of population-level parameters and individual planet properties.
- **Machine Learning Integration**: Can improve false positive rejection, detection sensitivity estimation, and population modeling.
- **Cross-survey Data Fusion**: Combining Kepler, TESS, radial velocity, and ground-based data will refine occurrence rates and habitable planet estimates.

8.3 Implications for Astrobiology

Statistical analysis of exoplanet populations informs the search for life beyond the Solar System. Accurate estimates of habitable planet frequencies guide target selection for follow-up characterization and the search for biosignatures. Expanding statistical methods to include planetary atmospheres and host star activity will enhance our understanding of planetary habitability.

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9. Conclusion

The statistical analysis of Kepler and TESS exoplanet datasets provides a comprehensive framework for understanding planetary populations and assessing habitability. By applying methods such as transit probability corrections, occurrence rate estimation, and probabilistic modeling, researchers have identified key patterns in planet sizes, orbital periods, multiplicity, and habitable zone distribution. Despite limitations arising from detection biases, stellar parameter uncertainties, and survey duration, these analyses have significantly advanced our knowledge of exoplanet demographics.

Future missions and improved statistical methodologies will further refine these insights, enabling more accurate predictions of potentially habitable worlds and informing the next generation of observational studies. The integration of large-scale statistical analysis with targeted follow-up observations represents a critical pathway toward answering fundamental questions about planetary formation, evolution, and the prevalence of habitable planets in our galaxy.

References (APA-style – Complete)

Berger, T. A., et al. (2020). Stellar parameters for exoplanet host stars from Gaia DR2. *The Astrophysical Journal Supplement Series*.

Borucki, W. J., et al. (2010). Kepler planet-detection mission: Introduction and first results. *Science*.

Christiansen, J. L., et al. (2015). Measuring transit signal recovery in the Kepler pipeline. *The Astrophysical Journal*.

Coughlin, J. L., et al. (2016). Planetary candidates observed by Kepler. *The Astrophysical Journal Supplement Series*.

Foreman-Mackey, D., et al. (2014). Exoplanet population inference and Bayesian methods. *The Astrophysical Journal*.

Fulton, B. J., et al. (2017). The California-Kepler Survey. III. A gap in the radius distribution of small planets. *The Astronomical Journal*.

Howard, A. W., et al. (2012). Planet occurrence within 0.25 AU of solar-type stars from Kepler. *The Astrophysical Journal Supplement Series*.

Kopparapu, R. K., et al. (2013). Habitable zones around main-sequence stars. *The Astrophysical Journal*.

Lissauer, J. J., et al. (2011). Architecture and dynamics of Kepler's multi-planet systems. *The Astrophysical Journal Supplement Series*.

Petigura, E. A., Howard, A. W., & Marcy, G. W. (2013). Prevalence of Earth-size planets orbiting Sun-like stars. *Proceedings of the National Academy of Sciences*.

Ricker, G. R., et al. (2015). Transiting Exoplanet Survey Satellite (TESS). *Journal of Astronomical Telescopes, Instruments, and Systems*.

Seager, S., & Mallén-Ornelas, G. (2003). A unique solution of planetary and stellar parameters from an extrasolar planet transit light curve. *The Astrophysical Journal*.